INCH-POUND

MIL-C-44112D
20 September 1993
SUPERSEDING
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18 April 1989

MILITARY SPECIFICATION

CRACKERS, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two crackers in a flexible bag for use as a component of Meal, Ready-To-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Cellulosic, Packaging
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurements

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN OIL CHEMISTS SOCIETY

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238	-	Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505	-	Density of Plastics by Density Gradient Technique
F 88	-	Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

ASSOCIATION OF VITAMIN CHEMISTS

Methods of Vitamin Assay

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(Application for copies should be addressed to the Interscience Publishers, a Division of John Wiley & Sons, 605 Third Avenue, New York, NY 10016.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Wheat flour. Wheat flour shall be of the commercial type known as cracker flour. It shall be bleached or unbleached and enriched or unenriched. If unenriched flour is used, equivalent enrichment required to conform to the definition and standard of identity for enriched flour (21 CFR Part 137), shall be incorporated by the manufacturer at the time of production of the finished product.

3.2.2 Shortening. The shortening shall be hydrogenated cottonseed, soybean, or peanut oil, or any combination thereof. The shortening shall contain antioxidants such as BHA, BHT, propyl gallate, or TBHQ singularly or in combination to the maximum total amount permitted by FDA regulations. The shortening shall conform to the following requirements when tested in accordance with the indicated methods of the Official and Tentative Methods of the American Oil Chemists Society.

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	Requirement	Method
Color (Lovibond)	2.5R maximum	Cc13b-45
Free fatty acid (FFA)	0.05 percent maximum	Ca5a-40
Fat stability (AOM)	100 hours (after antioxidant is added)	Cd12-57
Peroxide value (PV)	1.0 mEq oxygen/kg fat maximum (shall not exceed 60 days before use)	Cd8-53
Solid fat index (SFI)		Cd10-57
at 50°F	36 percent ± 5 percent	
at 70°F	24 percent ± 2 percent	
at 92°F	10 percent ± 5 percent	

3.2.3 Yeast, dried, inactive. The dried yeast shall be either primary grown, properly debittered brewers or torula yeast or a culture of Saccharomyces cerevisiae. It shall be mild in flavor and be finely powdered. Sufficient quantities of fortifying agents as determined by the Methods of Vitamin Assay of the Association of Vitamin Chemists shall be added to the yeast during processing to yield the following potencies:

Micrograms per gram

Thiamine	Not less than 1200
Riboflavin	Not less than 600
Niacin	Not less than 1000
Pyridoxine	Not less than 600

3.2.4 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.5 Leavening yeast. Leavening yeast shall be either compressed bakers or active dry yeast. The yeast shall be a culture of Saccharomyces cerevisiae.

3.2.6 Baking soda. Baking soda shall comply with the requirements of the Food Chemicals Codex. It shall be of a granulation commonly used in the cracker industry and shall be free flowing.

3.2.7 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.8 Malt preparation. Malt preparation shall be derived from water extract of barley malt, dried or evaporated, with or without cereal adjunct, or from malted barley flavor and a fine dextrose hydrate, with or without cereal adjunct.

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3.2.9 Protease enzyme. Protease enzyme shall be of high commercial grade and produced from either fungal or bacterial action.

3.2.10 Yeast food. Yeast food shall be an acid-type, mineral yeast food, dough improver and water conditioner. The yeast food shall provide nitrogen and mineral nutrition for the yeast, an oxidizing agent for proper, rapid dough conditioning, and calcium salts for water conditioning.

3.2.11 Calcium carbonate. Calcium carbonate shall comply with the requirements of the Food Chemicals Codex.

3.3 Preparation and processing. Processing shall be on a continuous process.

3.3.1 Cracker formulation. The crackers shall be prepared by the sponge-dough or straight dough method from ingredients in the following proportions:

Ingredients	Pounds
Wheat flour 1/	200
Shortening	28
Yeast, dried, inactive	3
Salt	1-1/2
Leavening yeast	As required
Baking soda	As required
Water	As required
Malt preparation, protease enzyme, yeast food (if necessary)	As required
Calcium carbonate ($CaCo_3$)	2/

1/ Ground crackers derived from clean but broken or otherwise defective crackers from previous runs of this formulation may be added to each batch in an amount not to exceed 5 percent of the flour ingredient.

2/ Add calcium carbonate at a rate to yield 2.5 grams per pound of finished crackers.

3.3.2 Cracker forming and baking (see 6.3). The dough shall be formed, laminated, cut into units, and uniformly well baked so as to yield 4 inch by 4 inch square crackers with a tolerance of ± 1/4 inch on any side. The thickness of the crackers shall be such that a stack of seven crackers (randomly selected and with every other cracker inverted when stacked) will be 1-1/2 inches, ± 1/4 inch in height. The height measurement shall be taken from the highest point of the stack. The number of baked crackers per pound shall not exceed 24. Each cracker shall have deep docker holes (visible holes extending completely through the baked cracker) placed regularly throughout its surface, approximately 1/4 inch apart. Each of the four sides of each cracker shall be scored once beginning on the outside edge (midpoint) and

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extending 1/2 inch in length toward the center of the cracker. Each cracker when broken along the partial scorelines shall form four 2 inch by 2 inch squares.

3.4 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, metal, or mold.

b. The crackers shall be uniformly well baked.

c. There shall be no color foreign to the product.

d. The crackers shall not be excessively brown in color (for example, dark brown, toasted, or scorched).

e. The crackers shall be clean, reasonably tender and crisp, and shall not be hard.

f. The crackers shall be free from any foreign or objectionable taste or odor (for example, musty, rancid, burnt, stale, alkaline, soapy, or excessively salty).

g. The crackers shall have clean cut edges and shall be reasonably free from dusting flour.

h. The average moisture content of the finished product shall be not more than 3.5 percent nor less than 2.0 percent.

i. No individual sample shall have a moisture content greater than 4.0 percent or less than 1.5 percent.

j. The pH (hydrogen-ion activity) of the finished cracker shall be not less than 6.5 nor greater than 8.0.

3.4.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110 "Current Good Manufacturing Practice in Manufacturing, Packing, or Holding Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.1.2 Material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements. In addition for the preformed bag, compliance to the requirements for inside bag dimensions and dimensions of manufacturer's seals may be verified by a certificate of conformance.

Material requirement	Requirement paragraphs	Test procedure
Thickness of films	5.1.1.1, 5.1.1.2, and 5.1.2.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378
Aluminum foil thickness	5.1.1.1. and 5.1.1.2	As specified in QQ-A-1876
Laminated material identification and construction	5.1.1.1 and 5.1.1.2	Laboratory evaluation
Color of laminated material	5.1.1.1 and 5.1.1.2	Visual evaluation

4.5.1.3 Unfilled preformed bag seal strength testing. The unfilled bags shall be tested for seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 inches or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 inch or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in bags. The sample unit shall be one unfilled bag. The sample size shall be the number of bags indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each bag in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement, or any seal failing to meet the average seal strength requirement specified in 5.1.1.1.1, shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.4 Preformed bag integrity and air tightness of manufacturer's seals testing. Bag integrity and air tightness shall be determined by placing crackers, fiberboard, or similar rigid material approximately 4 inches by 4 inches by 3/8 inch into the bag and sealing under a vacuum of not less than

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15 inches of mercury. Not less than 96 hours after vacuum sealing, the bag shall be examined for vacuum loss in accordance with the procedure in 4.5.4. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in units of unfilled preformed bags. The sample unit shall be one unfilled bag. The sample size shall be the number of bags indicated by inspection level I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65 for major defects.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, bag filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of formulation or other valid documents shall be cause for rejection of the involved product.

4.5.2.1 In-process examination at the packaging plant (other than manufacturing plant). Bulk-packed crackers which exceed 45 days from date of manufacture or are more than ten days from receipt shall be examined 72 hours or less before beginning of unit packing into two-cracker unit packs. Any sample unit with objectionable taste or odor (for example, musty, rancid, burnt, stale, alkaline, soapy) or any sample unit not reasonably tender and crisp shall be classified as a major defect. The lot size shall be expressed in cases. The sample unit shall be two crackers, randomly selected from each case. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects.

4.5.3 Filled and sealed laminated bag examination (unit packs shall be opened). The filled and sealed preformed and form-fill-seal laminated bags shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE I. Filled and sealed bag defects 1/

Category		Defect
Major	Minor	
101		Not clean 2/
102		Tear, hole, or open seal
103		Evidence of delamination 3/
104		Seal width less than 1/16 inch 4/
105		Bag has foreign odor

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TABLE I. Filled and sealed bag defects (cont'd) 1/

Category		Defect
Major	Minor	
106		One or more crushed crackers 5/
107		Presence of flex fractures in the aluminum foil 6/
108		Fewer than two crackers per bag
	201	Tear nick, tear notch, or serrations missing or does not facilitate easy opening
	202	One or more broken crackers 7/
	203	Seal width less than 1/8 inch, but greater than 1/16 inch
	204	Required labeling missing, incorrect, illegible, or that smudges
	205	Evidence of delamination 3/

1/ Any evidence of insect or rodent infestation shall be cause for rejection of the lot.

2/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause bag damage (for example, glass, metal filings) or generally detracts from the clean appearance of the package. The following examples shall not be classified as defects for unclean:

a. Foreign matter which presents no health hazard or potential bag damage and which can be readily removed by gently shaking the package, or by gently brushing the package with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one bag face (localized and aggregate).

c. Water spots.

3/ Delamination defect classification:

Major: Delamination of the outer ply in the bag seal area that can be propagated to expose aluminum foil at the food product edge of the bag after manual flexing of the delaminated area. To flex, the delaminated

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area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed 10 times by rotating both hands in alternating clockwise, counter-clockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the bag material. After flexing, the separated outer ply shall be grasped between thumb and forefinger and gently lifted toward the food product edge of the seal, or if the separated area is too small to be held between thumb and forefinger, a number of two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be classified as a major defect. Additionally, spot delamination of the outer ply in the body of the bag that is able to be propagated beyond its initial borders is also a major defect. To determine if the delaminated area is a defect, use the following procedure: Mark the outside edges of the delaminated area using a bold permanent marking pen. Open the bag and remove the contents. Cut the bag transversely not closer than 1/4 inch (± 1/16 inch) from the delaminated area. The bag shall be flexed in the area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be classified as a major defect.

Minor: Minor delamination of the outer ply in the bag seal area is acceptable and shall not be classified as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer ply delamination in the bag seal area or isolated spots of delamination in the body of the bag that do not propagate when flexed as described above shall be classified as minor defects.

4/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed bag.

5/ Crushed crackers is any unit where more than 1/8 of cracker volume is destroyed.

6/ To examine for flex fractures, the inside surface of each bag side shall be placed over a light source and the outside surface observed for the passage of light. Observation of light through the bag material in the form of a curved or straight line shall be evidence of the presence of flex fractures in the aluminum foil.

7/ A broken cracker is one that is broken into three or more pieces other than along cracker score lines.

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4.5.4 Laminated bag vacuum examination. Not less than 96 hours after filling and sealing, the filled and sealed bag shall be visually examined for conformance to the vacuum requirements in 5.1.1.1.2 or 5.1.1.2.1. The sealed bag shall continue to exhibit tight adherence to the large flat surfaces of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between the thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.5 Product examinations.

4.5.5.1 Product examination when unit packed in laminated bags as specified in 5.1.1 at the cracker manufacturer's plant only. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for minor defects.

4.5.5.2 Product examination when bulk packing as specified in 5.1.2.1 is utilized. The product shall be examined for the defects listed in table II. The lot size shall be expressed in shipping containers. The sample unit shall be one pound of crackers from each sample case. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for minor defects.

TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		Not uniformly well baked
102		Excessively brown in color (for example, dark brown, toasted, scorched)
103		Not reasonably tender and crisp 3/
104		Hard (for example, brittle, difficult to bite into) 3/
105		Objectionable taste or odor (for example, musty, rancid, burnt, stale, alkaline, soapy, excessively salty) 3/
106		More than 24 crackers per pound

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TABLE II. Product defects (cont'd) 1/ 2/

Category		Defect
Major	Minor	
107		Height of seven crackers greater than 1-3/4 inches 4/
	201	Edges not clean cut
	202	Not reasonably free from dusting flour (white powdery appearance on surface due to dusting powder)
	203	Length or width of cracker is less than 3-3/4 inches, or greater than 4-1/4 inches 3/
	204	Height of seven crackers less than 1-1/4 inch 4/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal, or mold) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot (see 3.4.1).

3/ Examine one cracker per sample unit.

4/ Select crackers from those used for defect 106.

4.5.6 Laminated bag closure seal testing. Eight filled and sealed bags shall be selected at random from the lot regardless of lot size and shall be tested in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 inches or 12 inches per minute. Machines that apply tensile load to the specimen by movement of the upper or lower clamp may be used. The test specimens shall be cut to a length suitable for proper mounting. For preformed bags, three adjacent specimens, 1/2 inch or 1 inch wide shall be cut from the closure seal of each bag in the sample. For the form-fill-seal bags, three adjacent specimens, 1/2 inch or 1 inch wide shall be cut from each side and each end of each bag in the sample. For the preformed bag, the average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form-fill-seal bag, the average seal strength of each side and end of the bag shall be calculated by averaging the test results of the three specimens cut from that

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side or end. The results shall be reported to the nearest 0.1 pound per inch of width. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.1.2 or 5.1.1.2.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Count per shipping container. Five shipping containers shall be selected at random from the lot. Any individual container which is more than 2 percent under the marked count shall be classified as a minor defect. Finding the average of the five shipping containers less than the average marked count shall be classified as a major defect. The finding of any defect shall be cause for rejection of the lot.

4.5.8 Product condition examination. Crackers bulk-packed for shipment to the unit packager shall be examined for the defects listed in table III. The lot size shall be expressed in shipping containers. The sample unit shall be three pounds of crackers from each shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects.

TABLE III. Cracker condition defects

Category	Defect
Major	
101	More than eight, 4 inch by 4 inch, crackers separated along a score line, broken, or crushed 1/
102	More than 14, 4 inch by 4 inch, crackers containing checking 2/

1/ A separated, broken, or crushed cracker is one that is in two or more distinct pieces, or has more than 1/16 of its area missing.

2/ Checking is the appearance of one or more fracture lines on the surface.

4.5.9 Shipping container external examination. Shipping containers bulk-packed for direct shipment to unit packagers shall be examined in accordance with the United States Standards for Condition of Food Containers.

4.5.10 Shipping container internal examination. Shipping containers bulk-packed for direct shipment to unit packagers shall be examined for the internal defects listed in table IV. The lot size shall be expressed in units of shipping containers. The sample unit shall be one filled shipping

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container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE IV. Shipping container internal defects

Category		Defect
Major	Minor	
101		Required polyethylene bag missing or has tear or hole
102		Corrugated glassine bag liner and/or dividers missing or not positioned as specified
103		Polyethylene bag does not completely cover top layer of crackers
104		Shipping container, polyethylene bag and/or corrugated glassine paper not clean
105		Corrugated box liner or top and bottom pads missing
106		Polyethylene bag and/or corrugated glassine bag liner has foreign odor
	201	Crackers not packaged in uniform rows and layers

4.5.11 Moisture content and pH testing. The finished product shall be analyzed for moisture content in accordance with the Official Methods of Analysis of the AOAC, method 925.09, and for pH concentration in accordance with the Official Methods of Analysis of the AOAC, method 943.02. The sample unit shall be one filled and sealed laminated bag or two crackers taken from different cartons or boxes, as applicable. The lot size shall be expressed in pounds. The sample size shall be the number of sample units indicated by inspection level S-2. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the moisture content or pH requirements in 3.4 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.12 Shipping container examination (for laminated barrier bags). Examination of the filled and closed shipping containers packed with two-cracker unit packs shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows:

Major: Markings missing, incorrect, or illegible.
Cushioning material missing or not as specified.

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Minor: Count not as specified (see 5.2.1.2).
More than 45 pounds packed in a box.
Bags not packed flat in layers.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.1).

5.1.1 Level A (for ration assembly). Two crackers shall be unit packed in flexible laminated barrier bags as specified in 5.1.1.1 or 5.1.1.2. Crackers shall be unit packed within 90 days of the date of manufacture.

5.1.1.1 Preformed bags. The preformed bags shall be fabricated from 0.002 inch thick (± 0.0002 inch) ionomer film bonded to 0.00035 inch thick (± 0.000035 inch) primed aluminum foil by melt extrusion with 0.001 (± 0.0002 inch) ionomer and subsequently bonding 0.0005 inch thick (± 0.00005 inch) polyethylene terephthalate film to the opposite side of the primed aluminum foil by melt extrusion with 0.0007 (± 0.00014 inch) low density polyethylene. Alternatively, 0.003 inch thick (± 0.0003 inch) linear low density polyethylene may be used in lieu of the ionomer. The ionomer film shall be the sodium neutralized copolymer of ethylene and methacrylic acid, having a melt index of 1.5 g/10 minutes in accordance with ASTM D 1238 and a density of 0.94 g/cc in accordance with ASTM D 1505. The ionomer used for bonding to the foil shall be the zinc neutralized copolymer of ethylene and methacrylic acid, having a melt index of 5.0 g/10 minutes and a density of 0.94 g/cc. When linear low density polyethylene is used, it shall be the copolymer of ethylene and octene 1, having a melt index range of 0.8 g/10 minutes to 1.2 g/10 minutes and a density range of 0.918 g/cc to 0.922 g/cc. The exterior (polyethylene terephthalate) surface of the bag shall be colored overall with a color in the range from 34079 through 34087, or 24052 through 24087, or 30045 through 30118 (except that 30109 may not be used) of FED-STD-595. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being unit packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into bags.

5.1.1.1.1 Bag construction. The bag shall be a flat style bag having inside dimensions of 5-1/2 inches by 6 inches (± 1/8 inch), with the 5-1/2 inch dimension being the opening of the bag. The empty bag shall be made by heat sealing all three edges with 3/8 inch (± 1/8 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals and preclude leakage of air into the unit pack through the seals, seal junctions or through defects created when the bags are filled and sealed as specified in 5.1.1.1.2. Integrity of the bag and air tightness of the seals shall be tested in accordance with 4.5.1.4. The side and bottom seals shall have an average seal strength of not less than 6 pounds per inch and no individual

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specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.5.1.3. A V-shaped or C-shaped (half round) tear nick or tear notch at least 1/32 inch deep, located 1 inch to 1-1/4 inches from the top edge of the bag, shall be made in one or both side seals. The distance between the inside edge of the tear nick or tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the bag may be provided with an extended or foldover lip, extended not more than 5/16 inch to facilitate opening and filling; if so, tear nick or tear notch location shall be measured from the top of the bag excluding the extended or foldover lip.

5.1.1.1.2 Bag filling and sealing. The crackers shall be filled into the bags in a manner to avoid puncturing the material. The filled bag shall be closed under a vacuum of not less than 15 inches or more than 20 inches of mercury (see 4.5.4) with a 3/8 ± 1/8 inch heat seal. If thermal impulse closure seals are used, a seal width of 1/8 inch will be acceptable. The closure seal shall be free of entrapped matter that reduces the effective closure seal to less than 1/16 inch wide. The closure seal shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch (see 4.5.6) and shall meet the requirements of the bag integrity test (see 4.5.1.4).

5.1.1.2 Horizontal form-fill-seal bags. The horizontal form-fill-seal bag shall consist of a formed tray-shaped body with a flat sheet, heat sealable cover, or a tray-shaped body with a tray-shaped heat sealable cover. The tray-shaped body, and the tray-shaped cover, shall be fabricated from a three-ply flexible laminate barrier material consisting of 0.0009 inch thick (± 0.00009 inch) oriented polypropylene bonded to 0.0007 inch thick (± 0.00007 inch) aluminum foil with 10 pounds per ream pigmented polyethylene or adhesive and bonding the opposite side of the aluminum foil to 0.003 inch thick (± 0.00006 inch) ionomer or a blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1 having a melt index range of 0.8 g/10 minutes to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 g/cc to 0.922 g/cc in accordance with ASTM D 1505. The heat sealable flat sheet cover shall be fabricated from material specified above except that the aluminum foil shall be a minimum 0.000315 inch thick. Alternatively, the tray-shaped body and flat sheet cover may be fabricated from material consisting of 0.0005 inch thick (± 0.00005 inch) polyester laminated to 0.0007 inch thick (± 0.00007 inch) aluminum foil that has 0.003 inch thick (± 0.0006 inch) ionomer, or polyethylene film in lieu of the above specified 0.0009 inch thick polypropylene. The polyester shall be on the outside of the package. The color requirements of the exterior (orientated polypropylene or polyester side) of the laminate shall be as specified in 5.1.1.1. The laminate shall be suitably formulated for food packaging and shall not impart any odor or flavor to the product being unit packed.

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5.1.1.2.1 Bag construction. The tray-shaped body and the tray-shaped cover shall be formed by drawing the flexible laminate material into an appropriately shaped cavity. The flat cover shall be in the form of a flat sheet of the barrier material taken from roll stock. Two crackers shall be placed into the tray-shaped body of the bag. The filled bag body shall then be hermetically sealed under a vacuum of not less than 15 inches nor more than 20 inches of mercury (see 4.5.4). The bag closure shall be effected by heat sealing the flat cover or the tray-shaped cover to the tray-shaped body along the entire bag perimeter. The outside dimensions of the sealed bags shall be 5-1/2 inches wide by 5-1/2 inches long (± 1/16 inch). The closure seal width shall be a minimum of 1/8 inch. The closure seal shall have an average seal strength of not less than 6 pounds per inch of width and no individual specimen shall have a seal strength of less than 5 pounds per inch (see 4.5.6). A V-shaped or C-shaped tear nick or tear notch shall be placed on one outside edge or two opposite outside edges of the bag to effect easy opening of the bag in the machine direction of the bag body and cover laminates. Each tear nick or tear notch shall be a minimum 1/32 inch deep, but shall not extend into the seal to a depth that would reduce the seal width to less than 1/8 inch. Alternatively, if the bag has serrated edges, the serrations may be used as tear notches provided that the serrations are located to effect easy opening in the machine direction of the bag laminates, the serrations are sharp (no plastic tailings exist), and the serration depth and the minimum seal width at the serrations are in accordance with the requirements specified for the V-shaped or C-shaped notches. The sealed bag shall not show any evidence of material degradation, aluminum stress cracking, delamination, or foreign odor when heat sealed or fabricated into bags. The seals shall be free of entrapped matter that reduces the effective closure seal to less than 1/16 inch wide.

5.1.2 Level B (for shipment to unit packager).

5.1.2.1 Bulk packing. The crackers shall be bulk packed in a polyethylene bag as specified in 5.1.2.1.1.

5.1.2.1.1 Bulk pack polyethylene bag. Approximately 18 pounds of crackers shall be unit packed in a polyethylene bag. The polyethylene bag shall be fabricated from 0.003 inch thick, unpigmented, food grade, high density polyethylene (specific gravity 0.941 - 0.965 g/cc). The crackers shall be positioned on edge and arranged in two layers of three rows each. All six rows shall be of equal length and there shall be no slack fill. Each layer and each row shall be separated by a single-faced corrugated glassine paper pad. The crackers, arranged as specified, shall be approximately 15-3/4 inches long, 12-5/8 inches wide, and 8-5/16 inches high. The crackers shall be separated from the inside surface of the bag by a U-shaped, single-faced corrugated glassine paper sheet, which completely covers the bottom surface and the two longest outside surface areas of the arranged crackers. The open end of the bag shall be located at the top of the unit pack. The bag shall be folded over the top of the crackers so that at least one layer of the polyethylene bag completely covers the top layer of crackers when the box flaps are closed. The polyethylene bag and the single-faced corrugated

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glassine paper sheet and pad material shall be suitably formulated for contact with food and shall not impart any odor or flavor to the crackers.

5.2 Packing. Packing shall be level C.

5.2.1 Level C.

5.2.1.1 Bulk pack shipping containers. Each 18 pound polyethylene bag of crackers, preserved as specified in 5.1.2.1.1, shall be packed in a 275 pound test, C-flute, RSC-L container, approximately 16 inches by 12-7/8 inches by 8-5/16 inches, with top and bottom corrugated pads. The box liner and the top and bottom pads shall be made from minimum 200 pound test corrugated fiberboard. The container shall be closed in accordance with PPP-B-636.

5.2.1.2 Laminated barrier bags. Ninety-six unit packs in accordance with 5.1.1 shall be packed flat in a snug-fitting fiberboard box constructed and closed in accordance with style RSC-L, type CF, class domestic, variety SW, grade 175 of PPP-B-636. Cushioning material, not less than 1/8 inch thick conforming to grade II, class B or C of A-A-1898 or cushioning material, not less than 1/16 inch thick conforming to type I of PPP-C-1797 shall be placed between each of the layers and at the bottom and the top of the box to minimize movement of individual unit packs.

5.3 Labeling and marking. Nutrition labeling shall be in accordance with the provisions of 21 CFR Part 101.9, Nutrition labeling of food.

5.3.1 Unit packs. Unit packs shall be permanently imprinted with the word "CRACKERS" with 3/8 inch high letters in accordance with MIL-STD-129.

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. The bulk pack shipping containers (see 5.2.1.1) shall have the date of cracker manufacture printed on one side and on one end of the boxes. In addition, the shipping containers shall be marked with the number of crackers per pound.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DOSISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.4.1 and 6.2).
e. Level of preservation required (see 5.1).
f. When bulk packing is required (see 5.1.2.1).

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6.2 First article. When a first article is required it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52-209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Cracker size, shape, height, and weight. To aid in maintaining the correct size and shape of crackers, as well as height and weight (see 3.3.2), and to hold "blistering" and "checking" to none or to a minimum, the following is listed for guidance:

a. Increase docker pins to forty or more per cup.
b. Enclose the cooling system.
c. Control cooling conditions from the oven to the packer to inhibit "checking" and "shattering."

6.4 Subject term (key word) listing.

Biscuit
Combat field feeding
Operational ration
Shelf stable

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8920-0553)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-44112D	1993 September 20

3. DOCUMENT TITLE

CRACKERS, FOR MEAL, READY-TO-EAT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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